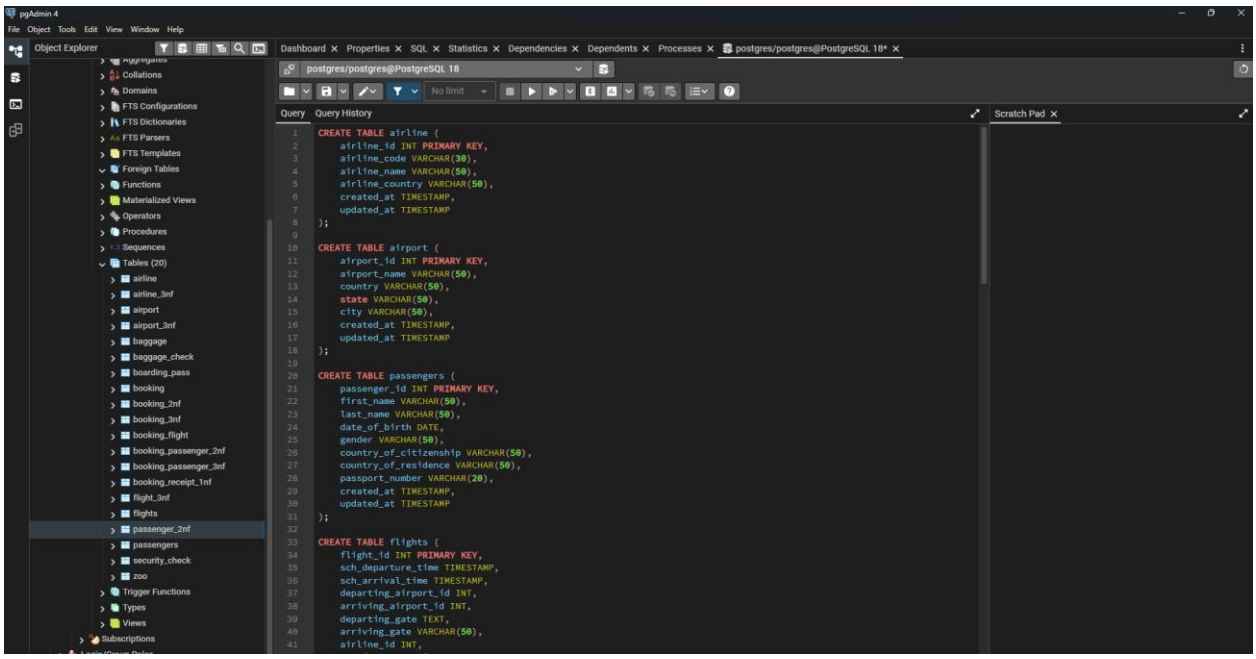
DML

1. Generate and insert 200 random rows in your airport database.
2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.
3. Update the airline country "KazAir" to "Turkey".
4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".
5. Delete all flights whose arrival in 2024 year.
6. Increase the price of all tickets in booking table for flights by 15%.
7. Delete all tickets where price is less than 10000.
8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).
9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.
10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.
11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.
12. Convert all country names in the airline table to uppercase to ensure data consistency.
13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date.
14. Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'
15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

First, I created a new tables as to not conflict with other tables from previous lab work – because I divided them into normalization steps, most of them don't contain all sufficient data:




```
    FOREIGN KEY (departing_airport_id)
        REFERENCES airport(airport_id),

    FOREIGN KEY (arriving_airport_id)
        REFERENCES airport(airport_id),

    FOREIGN KEY (airline_id)
        REFERENCES airline(airline_id)
);

CREATE TABLE booking (
    booking_id INT PRIMARY KEY,
    flight_id INT,
    passenger_id INT,
    booking_platform VARCHAR(50),
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    status VARCHAR(50),
    ticket_price DECIMAL(7,2),

    FOREIGN KEY (flight_id)
        REFERENCES flights(flight_id),

    FOREIGN KEY (passenger_id)
        REFERENCES passengers(passenger_id)
);

CREATE TABLE booking_flight (
    booking_flight_id INT PRIMARY KEY,
    booking_id INT,
    flight_id INT,
    created_at TIMESTAMP,
    updated_at TIMESTAMP,

    FOREIGN KEY (booking_id)
        REFERENCES booking(booking_id),

    FOREIGN KEY (flight_id)
        REFERENCES flights(flight_id)
);
```

```sql
CREATE TABLE baggage (
    baggage_id INT PRIMARY KEY,
    weight_in_kg DECIMAL(4,2),
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    booking_id INT,

    FOREIGN KEY (booking_id)
        REFERENCES booking(booking_id)
);

CREATE TABLE baggage_check (
    baggage_check_id INT PRIMARY KEY,
    check_result VARCHAR(50),
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    booking_id INT,
    passenger_id INT,

    FOREIGN KEY (booking_id)
        REFERENCES booking(booking_id),

    FOREIGN KEY (passenger_id)
        REFERENCES passengers(passenger_id)
);

CREATE TABLE boarding_pass (
    boarding_pass_id INT PRIMARY KEY,
    booking_id INT,
    seat VARCHAR(50),
    boarding_time TIMESTAMP,
    created_at TIMESTAMP,
    updated_at TIMESTAMP,

    FOREIGN KEY (booking_id)
        REFERENCES booking(booking_id)
);

CREATE TABLE security_check (
    security_check_id INT PRIMARY KEY,
    check_result VARCHAR(20),
```

Only after that I went ahead with tasks

1. Generate and insert 200 random rows in your airport database.

Query Query History

```
INSERT INTO airport (airport_id, airport_name, country, state, city, created_at, updated_at)
SELECT
    5000 + gs,
    'Airport ' || gs,
    CASE WHEN random() < 0.5 THEN 'Kazakhstan' ELSE 'France' END,
    NULL,
    CASE WHEN random() < 0.5 THEN 'Almaty' ELSE 'Paris' END,
    CURRENT_TIMESTAMP,
    CURRENT_TIMESTAMP
FROM generate_series(1, 200) AS gs;
```

Data Output Messages Notifications

Showing rows: 1 to 200

	airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	5001	Airport 1	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
2	5002	Airport 2	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
3	5003	Airport 3	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
4	5004	Airport 4	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
5	5005	Airport 5	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
6	5006	Airport 6	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
7	5007	Airport 7	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
8	5008	Airport 8	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
9	5009	Airport 9	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
10	5010	Airport 10	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
11	5011	Airport 11	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
12	5012	Airport 12	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
13	5013	Airport 13	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
14	5014	Airport 14	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
15	5015	Airport 15	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
16	5016	Airport 16	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
17	5017	Airport 17	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
18	5018	Airport 18	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
19	5019	Airport 19	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
20	5020	Airport 20	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747

Total rows: 200 Query complete 00:00:00.138

I used CASE WHEN to randomize the selection between France and Kazakhstan as a choose.

2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table:

Query Query History

```
INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
VALUES (101, NULL, 'KazAir', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	[null]	KazAir	Kazakhstan	2026-09-27 17:08:48.291306	2026-09-27 17:08:48.291306

3. Update the airline country "KazAir" to "Turkey"(I used UPDATE, SET, WHERE)

Query Query History

```sql
UPDATE airline
SET airline_country = 'Turkey', updated_at = CURRENT_TIMESTAMP
WHERE airline_name = 'KazAir';
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	[null]	KazAir	Turkey	2026-09-27 17:08:48.291306	2026-09-27 17:09:49.936158

4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland":

Query Query History

```sql
INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
VALUES
  (102, NULL, 'AirEasy', 'France', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
  (103, NULL, 'FlyHigh', 'Brazil', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
  (104, NULL, 'FlyFly',  'Poland', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	[null]	KazAir	Turkey	2026-09-27 17:08:48.291306	2026-09-27 17:09:49.936158
2	102	[null]	AirEasy	France	2026-09-27 17:11:11.289873	2026-09-27 17:11:11.289873
3	103	[null]	FlyHigh	Brazil	2026-09-27 17:11:11.289873	2026-09-27 17:11:11.289873
4	104	[null]	FlyFly	Poland	2026-09-27 17:11:11.289873	2026-09-27 17:11:11.289873

By using INSERT INTO and VALUES syntaxes

5. Delete all flights whose arrival in 2024 year:

Query Query History Scratch Pad

```sql
DELETE FROM flights
WHERE sch_arrival_time >= '2024-01-01'
  AND sch_arrival_time <  '2025-01-01';
```

Data Output Messages Notifications

	flight_id [PK] integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departure_time timestamp without time zone	act_arrival_time timestamp without time zone

Here I made a mistake, took a screenshot after running the query, should have shown before and after.

6. Increase the price of all tickets in booking table for flights by 15%:

Before:

Query Query History

```
SELECT * FROM booking
```

Data Output Messages Notifications

Showing rows: 1 to 1

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	1	1	1	FREEDOM	2026-09-27 17:21:16.509646	2026-09-27 17:21:16.509646	Paid	300.00

After:

Query Query History

```
UPDATE booking
SET ticket_price = ticket_price * 1.15,
    updated_at = CURRENT_TIMESTAMP;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	1	1	1	FREEDOM	2026-09-27 17:21:16.509646	2026-09-27 17:22:04.750044	Paid	345.00

(ticket_price went from 300 to 345)

7. Delete all tickets where price is less than 10000:

Before:

Query Query History

```
SELECT * FROM booking
```

Data Output Messages Notifications

Showing rows: 1 to 2

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	1	1	1	FREEDOM	2026-09-27 17:21:16.509646	2026-09-27 17:22:04.750044	Paid	345.00
2	2	2	2	FREEDOM	2026-09-27 17:24:31.803493	2026-09-27 17:24:31.803493	Paid	15000.00

After:

Query Query History

```
DELETE FROM booking
WHERE ticket_price < 10000;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	2	2	2	FREEDOM	2026-09-27 17:24:31.803493	2026-09-27 17:24:31.803493	Paid	15000.00

8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL):

Before

Query Query History

```
SELECT * FROM airline
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	[null]	KazAir	Turkey	2026-09-27 17:08:48.291306	2026-09-27 17:09:49.936158
2	102	[null]	AirEasy	France	2026-09-27 17:11:11.289873	2026-09-27 17:11:11.289873
3	103	[null]	FlyHigh	Brazil	2026-09-27 17:11:11.289873	2026-09-27 17:11:11.289873
4	104	[null]	FlyFly	Poland	2026-09-27 17:11:11.289873	2026-09-27 17:11:11.289873

After

Query Query History

```
UPDATE airline
SET airline_code = 'UNK', updated_at = CURRENT_TIMESTAMP
WHERE airline_code IS NULL;
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	UNK	KazAir	Turkey	2026-09-27 17:08:48.291306	2026-09-27 17:27:46.900552
2	102	UNK	AirEasy	France	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
3	103	UNK	FlyHigh	Brazil	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
4	104	UNK	FlyFly	Poland	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552

9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked':

Query Query History

```
SELECT * FROM baggage_check
```

Data Output Messages Notifications

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	2	checked	2026-09-27 17:34:09.064634	2026-09-27 17:34:09.064634	2	2

After:

Query Query History

```
DELETE FROM baggage_check
WHERE created_at < '2023-06-01'
  AND check_result = 'Not checked';
```

Data Output Messages Notifications

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	2	checked	2026-09-27 17:34:09.064634	2026-09-27 17:34:09.064634	2	2

(since there are no older baggage checks, nothing is changed)

10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh’

Query Query History

```
DELETE FROM airport
WHERE state IS NULL
  AND (city = 'Mlawe' OR city = 'Kepuh');
```

Data Output Messages Notifications

Showing rows: 1 to 200

	airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	5001	Airport 1	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
2	5002	Airport 2	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
3	5003	Airport 3	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
4	5004	Airport 4	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
5	5005	Airport 5	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
6	5006	Airport 6	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
7	5007	Airport 7	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
8	5008	Airport 8	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
9	5009	Airport 9	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
10	5010	Airport 10	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
11	5011	Airport 11	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
12	5012	Airport 12	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
13	5013	Airport 13	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
14	5014	Airport 14	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
15	5015	Airport 15	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
16	5016	Airport 16	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
17	5017	Airport 17	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
18	5018	Airport 18	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
19	5019	Airport 19	Kazakhstan	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
20	5020	Airport 20	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
21	5021	Airport 21	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
22	5022	Airport 22	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
23	5023	Airport 23	France	[null]	Almaty	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
24	5024	Airport 24	France	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747
25	5025	Airport 25	Kazakhstan	[null]	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747

11.Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp:

Query Query History

```
SELECT * FROM baggage_check
```

Data Output Messages Notifications

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	2	checked	2026-09-27 17:34:09.064634	2026-09-27 17:34:09.064634	2	2

After

Query Query History

```sql
INSERT INTO baggage_check (baggage_check_id, check_result, created_at, updated_at, booking_id, passenger_id)
VALUES (9001, 'Not checked', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 2, 1)
RETURNING baggage_check_id, created_at;
```

Data Output Messages Notifications

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	2	checked	2026-09-27 17:34:09.064634	2026-09-27 17:34:09.064634	2	2
2	9001	Not checked	2026-09-27 17:41:04.694352	2026-09-27 17:41:04.694352	2	1

12. Convert all country names in the airline table to uppercase to ensure data consistency:

Query Query History

```sql
UPDATE airline
SET airline_country = UPPER(airline_country);
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	UNK	KazAir	TURKEY	2026-09-27 17:08:48.291306	2026-09-27 17:27:46.900552
2	102	UNK	AirEasy	FRANCE	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
3	103	UNK	FlyHigh	BRAZIL	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
4	104	UNK	FlyFly	POLAND	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552

13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date

Query Query History

```sql
SELECT * FROM airline
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	UNK	KazAir	TURKEY	2026-09-27 17:08:48.291306	2026-09-27 17:27:46.900552
2	102	UNK	AirEasy	FRANCE	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
3	103	UNK	FlyHigh	BRAZIL	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
4	5	UNK	FlyFly	POLAND	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552

After

Query Query History

```sql
UPDATE airline
SET airline_name = 'Global Airways',
    airline_country = 'United Kingdom',
    updated_at = CURRENT_DATE
WHERE airline_id = 5;
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	UNK	KazAir	TURKEY	2026-09-27 17:08:48.291306	2026-09-27 17:27:46.900552
2	102	UNK	AirEasy	FRANCE	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
3	103	UNK	FlyHigh	BRAZIL	2026-09-27 17:11:11.289873	2026-09-27 17:27:46.900552
4	5	UNK	Global Airways	United Kingdom	2026-09-27 17:11:11.289873	2026-09-27 00:00:00

14. Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'

Query Query History

```sql
UPDATE airport
SET state = 'Capital District'
WHERE city = 'Astana' OR city = 'London' OR city = 'Tokyo';
```

Data Output Messages Notifications

Showing rows: 1 to 200

	airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
93	5004	Airport 4	France	Capital District	Paris	2026-09-27 17:06:39.634747	2026-09-27 17:06:39.634747

15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked')

Query Query History

```sql
UPDATE baggage_check
SET check_result = 'Checked', updated_at = CURRENT_TIMESTAMP
WHERE check_result = 'Not checked'
  AND created_at >= '2024-03-01'
  AND created_at <  '2024-04-01';
```

Data Output Messages Notifications

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	2	checked	2026-09-27 17:34:09.064634	2026-09-27 17:34:09.064634	2	2
2	9001	Not checked	2026-09-27 17:41:04.694352	2026-09-27 17:41:04.694352	2	1